

# CM0246 Discrete Structures

## Relations and Their Properties

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# Preliminaries

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## Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Relations

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Recall the definition of Cartesian product

Let  $A$  and  $B$  be sets. The Cartesian product of  $A$  and  $B$  is

$$A \times B = \{ (a, b) \mid a \in A \wedge b \in B \}.$$

Example

Let  $A = \{a, b\}$  and  $B = \{1, 2\}$ . Then

$$A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2)\}.$$

# Relations

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## Definition

Let  $A$  and  $B$  be sets. A **binary relation** from  $A$  to  $B$  is a subset of  $A \times B$ .

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## Example (trivial relations)

$R = \emptyset$  and  $R = A \times B$  are relations from  $A$  to  $B$ .

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## Example

See whiteboard.

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## Example

See whiteboard.

## Notation

We shall use  $(a, b) \in R$  and  $a R b$ .

The slides for the 6th ed. of Rosen's textbook use  $\langle a, b \rangle$ .

# Relations

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## Relations and functions

The functions are relations with additional constraints.



# Relations

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## Definition

A **relation on** a set  $A$  is a relation from  $A$  to  $A$ .

# Relations

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## Example

Some relations on  $\mathbb{Z}$ :

$$R_1 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \leq b \},$$

$$R_2 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a > b \},$$

$$R_3 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a = b \vee a = -b \},$$

$$R_4 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a = b \},$$

$$R_5 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a = b + 1 \},$$

$$R_6 = \{ (a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a + b \leq 3 \}.$$

# Properties of Relations

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## Definition

Let  $R$  be a relation on a set  $A$ . The relation  $R$  is

**reflexive** iff  $\forall x(xRx)$ ,

**symmetric** iff  $\forall x\forall y(xRy \rightarrow yRx)$ ,

**antisymmetric** iff  $\forall x\forall y((xRy \wedge yRx) \rightarrow x = y)$  and

**transitive** iff  $\forall x\forall y\forall z((xRy \wedge yRz) \rightarrow xRz)$ .

# Properties of Relations

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## Example

See slides § 8.1, p. 5 for the 6th ed. of Rosen's textbook.

# Combing Relations

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See slides § 8.1, pp. 6-8 for the 6th ed. of Rosen's textbook.

# Combing Relations

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## Definition

Let  $R$  be a relation from  $A$  to  $B$  and let  $S$  be a relation from  $B$  to  $C$ .

The **composition** of  $S$  with  $R$ , denoted  $S \circ R$ , is the relation from  $A$  to  $C$  where if  $(a, b) \in R$  and  $(b, c) \in S$  then  $(a, c) \in S \circ R$ .

# Combing Relations

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## Example (composition of relations)

Let  $A = \{1, 2, 3\}$ ,  $B = \{1, 2, 3, 4\}$  and  $C = \{0, 1, 2\}$ .

Let  $R$  and  $S$  be the relations from  $A$  to  $B$  and  $B$  to  $C$ , respectively, given by

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\},$$

$$S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\},$$

then  $S \circ R$  is the relation from  $A$  to  $C$ , given by

$$S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}.$$

# Combing Relations

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## Problem 31 (p. 448)

Let  $R$  be the relation on the set of people consisting of pairs  $(a, b)$ , where  $a$  is a parent of  $b$ .  
Let  $S$  be the relation on the set of people consisting of pairs  $(a, b)$ , where  $a$  and  $b$  are siblings (brothers or sisters).



# Combing Relations

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What are  $S \circ R$  and  $R \circ S$ ?

# Combing Relations

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What are  $S \circ R$  and  $R \circ S$ ?

- $(a, b) \in S \circ R$  if exists  $c$  such that  $(a, c) \in R$  ( $a$  is parent of  $c$ ) and  $(c, b) \in S$  ( $c$  is sibling of  $b$ ), that is

$$S \circ R = \{ (a, b) \mid a \text{ is a parent of } b \text{ and } b \text{ has a sibling} \}.$$

# Combing Relations

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$$S \circ R = \{ (a, b) \mid a \text{ is a parent of } b \text{ and } b \text{ has a sibling} \}.$$

- $(a, b) \in R \circ S$  if exists  $c$  such  $(a, c) \in S$  ( $a$  is sibling of  $c$ ) and  $(c, b) \in R$  ( $c$  is parent of  $b$ ), that is

$$R \circ S = \{ (a, b) \mid a \text{ is an aunt or uncle of } b \}.$$

# Combing Relations

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## Definition

Let  $R$  be a relation on the set  $A$ . The **powers**  $R^n$ , for  $n \in \mathbb{Z}^+$  are defined recursively by

$$\begin{aligned} R^1 &= R, \\ R^{n+1} &= R^n \circ R. \end{aligned}$$

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## Example

See slides § 8.1, pp. 9-10 for the 6th ed. of Rosen's textbook.

# Combing Relations

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## Theorem 1 (p. 446)

Let  $R$  be a relation on a set  $A$ . The relation  $R$  is transitive iff  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$ .

(continued on next slide)

# Combing Relations

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Proof of  $\Rightarrow$  (if  $R$  is transitive implies  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$ ).

By induction on  $n \in \mathbb{Z}^+$ .

# Combing Relations

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By induction on  $n \in \mathbb{Z}^+$ .

1.  $P(n)$ : if  $R$  is transitive implies  $R^n \subseteq R$ .



# Combing Relations

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2. Basis step  $P(1)$ :  $R^1 = R \subseteq R$

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Inductive hypothesis  $P(k)$ : if  $R$  is transitive implies  $R^k \subseteq R$

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Inductive hypothesis  $P(k)$ : if  $R$  is transitive implies  $R^k \subseteq R$   
Let's prove  $P(k + 1)$ :

# Combing Relations

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By induction on  $n \in \mathbb{Z}^+$ .

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Let's prove  $P(k+1)$ :

3.1 Let  $(a, b) \in R^{k+1}$ .

# Combing Relations

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By induction on  $n \in \mathbb{Z}^+$ .

1.  $P(n)$ : if  $R$  is transitive implies  $R^n \subseteq R$ .

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Inductive hypothesis  $P(k)$ : if  $R$  is transitive implies  $R^k \subseteq R$

Let's prove  $P(k+1)$ :

3.1 Let  $(a, b) \in R^{k+1}$ .

3.2 Exists  $x \in A$  such that  $(a, x) \in R^k$  and  $(x, b) \in R$  (definition of  $\circ$ ).

# Combing Relations

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Proof of  $\Rightarrow$  (if  $R$  is transitive implies  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$ ).

By induction on  $n \in \mathbb{Z}^+$ .

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Let's prove  $P(k+1)$ :

3.1 Let  $(a, b) \in R^{k+1}$ .

3.2 Exists  $x \in A$  such that  $(a, x) \in R^k$  and  $(x, b) \in R$  (definition of  $\circ$ ).

3.3  $(a, x) \in R$  (IH).

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Proof of  $\Rightarrow$  (if  $R$  is transitive implies  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$ ).

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Let's prove  $P(k+1)$ :

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3.2 Exists  $x \in A$  such that  $(a, x) \in R^k$  and  $(x, b) \in R$  (definition of  $\circ$ ).

3.3  $(a, x) \in R$  (IH).

3.4 If  $(a, x) \in R$  and  $(x, b) \in R$  then  $(a, b) \in R$  ( $R$  is transitive).

# Combing Relations

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Proof of  $\Rightarrow$  (if  $R$  is transitive implies  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$ ).

By induction on  $n \in \mathbb{Z}^+$ .

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3.2 Exists  $x \in A$  such that  $(a, x) \in R^k$  and  $(x, b) \in R$  (definition of  $\circ$ ).

3.3  $(a, x) \in R$  (IH).

3.4 If  $(a, x) \in R$  and  $(x, b) \in R$  then  $(a, b) \in R$  ( $R$  is transitive).

3.5  $R^{k+1} \subseteq R$ . ■

(continued on next slide)



# Combing Relations

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Proof of  $\Leftarrow$  (if  $R^n \subseteq R$  for  $n \in \mathbb{Z}^+$  implies  $R$  is transitive).

1 Suppose that  $(a, b) \in R$  and  $(b, c) \in R$ .

2  $(a, c) \in R^2$ .

(def. of  $R^2$ )

3  $(a, c) \in R$ .

( $R^2 \subseteq R$ )

4 Therefore,  $R$  is transitive. ■

# Inverse and Complementary Relations

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## Definition

Let  $R$  be a relation from  $A$  to  $B$ . The **inverse relation** from  $B$  to  $A$ , denoted by  $R^{-1}$ , is the set of ordered pairs

$$R^{-1} = \{ (b, a) \in B \times A \mid (a, b) \in R \}.$$

# Inverse and Complementary Relations

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## Definition

Let  $R$  be a relation from  $A$  to  $B$ . The **complementary relation** from  $A$  to  $B$ , denoted by  $\overline{R}$ , is the set of ordered pairs

$$\overline{R} = \{ (a, b) \in A \times B \mid (a, b) \notin R \}.$$

# References

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Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).